

LLM Model Choice: `openai/gpt-oss-120b`

Status: Active

Decision date: 2 August 2026

Applies to: WhatsApp intake agent

1. Summary

The WhatsApp intake agent runs on **`openai/gpt-oss-120b`**, an open-weight 117B-parameter Mixture-of-Experts model from OpenAI, served through **Groq** (primary, when its paid tier is available) and **OpenRouter** (currently primary).

The decision rests on four requirements specific to this system, in priority order:

1. **Reliable multi-step tool calling** — the agent has 12 tools and must chain them
2. **Multilingual** — English, Hindi and Marathi in the same conversation
3. **Low cost** — a single-location service business, not a funded product
4. **No vendor lock-in** — the model must be available from more than one provider

Requirement 4 is the one that separates this model from every closed alternative, and it has already paid for itself once (see section 5).

2. What the model actually has to do

This is not a chatbot. Per one customer message the model must:

- Decide which of five flows applies (new request / status / FAQ / complaint / reschedule-cancel)
- Call the right tools from a set of 12
- Chain them correctly — e.g. `identify_customer` → `create_or_get_request` → `update_request` → `submit_request`
- Follow a ~175-line system prompt with strict negative constraints ("never invent a technician name", "never quote a price", "never re-ask for a saved field")
- Reply in the customer's language, in short operational Indian English

A model that is merely *fluent* fails here. The binding constraint is **function-calling accuracy under a long instruction set**, not prose quality.

3. Why gpt-oss-120b

3.1 Tool calling is a first-class capability

The model was built for agentic use — it supports **native tool use, function calling, structured output, and configurable reasoning depth**, and demonstrates strong function-calling performance on the **Tau-Bench Retail** benchmark, which specifically measures multi-turn tool use in a customer-service setting. That is close to a direct analogue of this workload.

Fireworks' evaluation describes both gpt-oss models as *"very strong reasoning models that excel at problem solving and tool calling."*

3.2 Reasoning quality is well above its price class

- **90% on MMLU** (general knowledge and reasoning)
- **Surpasses OpenAI o3-mini and approaches o4-mini** on accuracy

We are paying open-model prices for near-o4-mini instruction-following. That matters because our system prompt is long and rule-dense — a weaker model drifts and starts inventing prices or re-asking for the customer's name.

3.3 Context window is sufficient

131K tokens. Our worst-case payload is roughly 4,300 tokens (system prompt + 12 tool definitions) plus 12 turns of history — under 10K. We are using **less than 8%** of the window, so there is no truncation risk and plenty of headroom if the prompt grows.

3.4 Multilingual

Handles English, Hindi and Marathi in the same conversation, including Devanagari input, which the system prompt explicitly requires.

4. Cost comparison — using our real numbers

Measured from the live agent loop:

Metric	Value
Fixed payload per LLM call	~4,300 tokens (2,700 prompt + 1,600 tool defs)
LLM calls per customer message	up to 3
Messages per completed request	~5
Input tokens per completed request	~90,000
Output tokens per completed request	~3,000

At **1,000 completed requests/month** = 90M input + 3M output tokens:

Model	Input \$/M	Output \$/M	Monthly cost	vs. our choice
gpt-oss-120b (OpenRouter)	\$0.03	\$0.17	~\$3.21	—
GPT-4o-mini	\$0.15	\$0.60	~\$15.30	4.8x
Gemini 2.5 Flash	\$0.30	\$2.50	~\$34.50	10.7x
Claude Haiku 4.5	\$1.00	\$5.00	~\$105.00	32.7x

All four prices are vendor-published (sources in section 8). The spread is wider than the headline per-token numbers suggest because intake is **input-heavy** — a long prompt and a short reply — and that is exactly where the gap is largest. Independent analysis puts gpt-oss-120b at ~4.7x cheaper than Gemini 2.5 Flash on a blended 3:1 mix; on our 30:1 mix it is ~10.7x.

On capability, the gap does not justify the spread. Against Gemini 2.5 Flash the published comparison is "*both models are evenly matched across the benchmarks*" — its wins are a 1M context window and multimodal input, and we use 8% of a 131K window and handle images on a separate path. Neither applies to us.

ON CLAUDE HAIKU 4.5 — REJECTED ON COST, NOT ON QUALITY

Haiku 4.5 is the **strongest of the alternatives on capability**: excellent multi-step tool calling and strong Hindi/Marathi handling — our top two requirements — and its 200K context is more than sufficient. It is rejected on section 5 alone: closed weights, single vendor, and 32x the cost for a capability advantage we cannot measure on a 12-tool WhatsApp intake flow.

5. The decisive factor: no vendor lock-in

`gpt-oss-120b` is **open-weight**, so it is served by many independent providers — Groq, OpenRouter (17+ upstream providers), Fireworks, DeepInfra, and others. Prices vary slightly by provider (\$0.03–\$0.04 input, \$0.17–\$0.19 output), which means we can shop on price and latency without changing the model.

THIS IS NOT THEORETICAL — IT HAPPENED ON 2 AUGUST 2026

Groq's paid Developer tier was closed to new signups ("temporarily unavailable due to high demand"), leaving us stuck on the free 8,000 tokens/minute tier — a tier our ~15–20K-tokens-per-message workload structurally cannot fit inside. Every message was hitting a rate limit and falling into a retry path that took **minutes**.

Because the model is multi-provider, the fix was:

```
LLM_PRIMARY=openrouter  
OPENROUTER_MODEL=openai/gpt-oss-120b
```

Two environment variables. Same model, same prompt, same 12 tool definitions, same behaviour, zero code changes, zero re-testing of agent logic.

Had we built on a closed model, that outage would have meant a provider migration: new SDK, re-tuned prompt, re-validated tool-calling behaviour, and days of work — in the middle of a live customer-facing outage.

We are buying a capability, not a vendor. That optionality is the single strongest argument for this choice, and it is only available with an open-weight model.

6. Alternatives considered and rejected

Option	Why rejected
gpt-oss-20b	Same family, weaker reasoning. Was our OpenRouter fallback on the <code>:free</code> route; the free tier is queued and slow, and the smaller model was less reliable at chaining our 12 tools. Kept only as an emergency fallback.
Claude Haiku 4.5	The best alternative on merit — excellent tool calling, strong Hindi/Marathi, 200K context. Rejected on cost and lock-in only: 32.7x our spend (\$105/mo vs \$3.21), closed weights, single vendor. Revisit if intake accuracy ever becomes the binding constraint instead of cost.
Gemini 2.5 Flash	10.7x our cost, evenly matched on benchmarks, single-vendor. Its wins (1M context, multimodal) are irrelevant to us.
GPT-4o-mini	4.8x cost, closed weights, single vendor. The cheapest closed option, with no tool-calling advantage at this level — and still fails requirement 4.
Self-hosting gpt-oss-120b	A 117B MoE model needs serious GPU hardware. Nonsensical at ~\$3/month of API spend.

7. Current configuration

Variable	Value	Notes
<code>GROQ_MODEL</code>	<code>openai/gpt-oss-120b</code>	Fallback while Groq is on the free tier
<code>OPENROUTER_MODEL</code>	<code>openai/gpt-oss-120b</code>	Paid route — no <code>:free</code> suffix
<code>LLM_PRIMARY</code>	<code>openrouter</code>	Flip to <code>groq</code> when the Developer tier reopens

Review trigger: re-open this decision when Groq's Developer tier becomes available (Groq is meaningfully faster than most OpenRouter routes), or if monthly spend exceeds \$25 — which at current rates means traffic has grown ~8x and the token-trimming work in the agent prompt should be done first.

8. Sources

- OpenRouter — gpt-oss-120b pricing & benchmarks — <https://openrouter.ai/openai/gpt-oss-120b>
- Fireworks AI — OpenAI gpt-oss 20b & 120b: overview & benchmarking — <https://fireworks.ai/blog/openai-gpt-oss>
- SmythOS — gpt-oss-120b and 20b: speed, accuracy, real results — <https://smythos.com/developers/ai-models/openai-gpt-oss-120b-and-20b-speed-accuracy-and-real-results/>
- LLM-Stats — Gemini 2.5 Flash vs GPT-OSS-120B — <https://llm-stats.com/models/compare/gemini-2.5-flash-vs-gpt-oss-120b>
- GPT-OSS-120B provider pricing comparison — <https://pricepertoken.com/pricing-page/model/openai-gpt-oss-120b>
- Groq — rate limits — <https://console.groq.com/docs/rate-limits>

7. OpenAI — GPT-4o mini pricing announcement (\$0.15 / \$0.60) — <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>
8. OpenAI — API pricing — <https://developers.openai.com/api/docs/pricing>
9. Anthropic — Claude model pricing (\$1.00 / \$5.00 for Claude Haiku 4.5, 200K context)